Bachelor of Technology

Internship/Project
Final Presentation

Academic Year 2026–27

Semester VIII



SVKM's NMIMS

Mukesh Patel School of Technology Management and Engineering, Vile Parle, Mumbai- 400056

Department of Artificial Intelligence

A PRESENTATION ON

SELF-DRIVING MINI CAR
Lane Detection, Obstacle Avoidance and Traffic Sign Recognition

By

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Under the guidance of

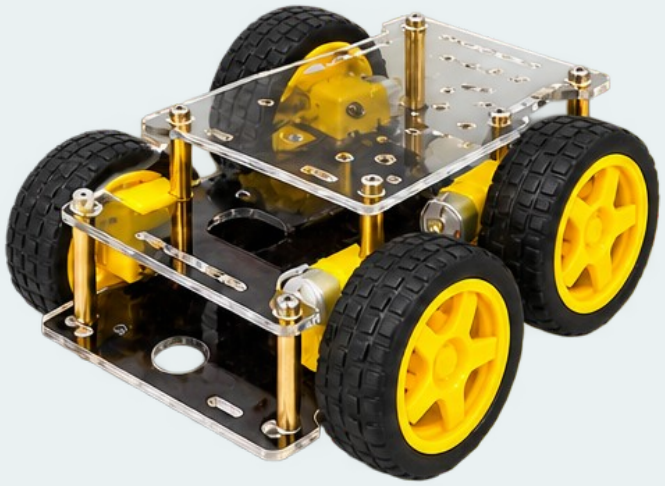
Dr. Hima Deepthi Vankayalapati

Department of Artificial Intelligence

PERCEPTION | DECISION | CONTROL

Introduction and Problem Statement

THE PROJECT AT A GLANCE / SEE THE TRACK • CHOOSE AN ACTION • MOVE



A small autonomy testbed

Concept illustration • skid-steer drive

How do we turn what the camera sees into useful motion?



Perceive

Read lanes, signs and visual hazards.



Decide

Resolve instructions and safety priorities.



Act

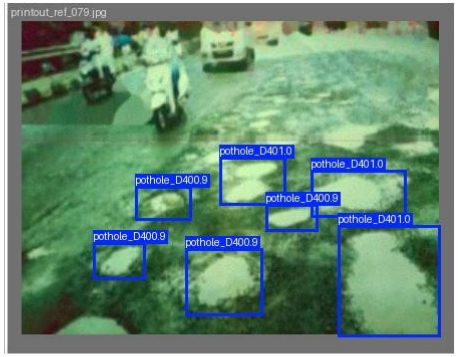
Execute motor commands and report state.

Tools: Python / OpenCV • YOLO / NCNN • Picamera2 / pyserial • Uno C++

Sources: final report Ch. 1/3; Krishna main 9aeb238. Hardware shown here is conceptual illustration, not prototype photography.

Literature Review and Research Gap

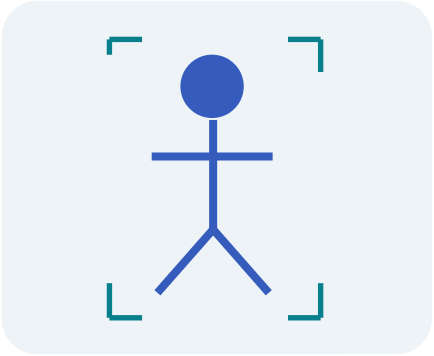
WHAT PRIOR WORK ENABLES / DETECTION • TRACKING • ACCESSIBLE ROBOTICS



Learned detection

YOLOv10 + RDD2022 [1,3]

Use compact detectors; check the data domain.



Efficient tracking

KCF owner tracking [2]

Track between detections; recover after target loss.



Accessible robotics

OpenBot + Duckietown [4,5]

Keep the platform modular and the workflow reproducible.

Project contribution: connect these methods through clear decisions and embedded execution.

Sources: [1–5] on slide 16. Detection image: repository validation crop. Person and board visuals are conceptual illustrations.

System Architecture / Block Diagram

ILLUSTRATED ARCHITECTURE / FOLLOW THE CAMERA-TO-MOTOR PATH

PERCEPTION / DECISION

EMBEDDED CONTROL

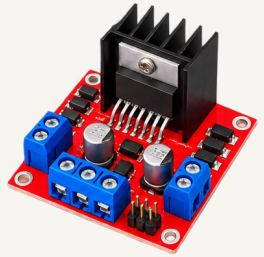




Serial

state





CSI camera

Live frames

Raspberry Pi

See + decide

Arduino Uno

Sense + execute

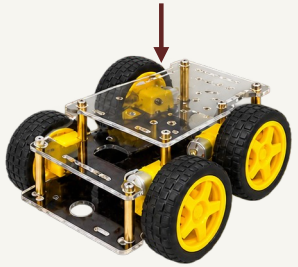
L298N driver

Motor power

Dashboard
Live view + status

Front sonar feeds Uno





Commands travel to Uno; telemetry returns to Pi. Sonar is read on Uno, not through Pi GPIO.

Source: Krishna main 9aeb238 runtime and firmware. Component art is illustrative; arrows show functional interfaces, not physical wiring.

Methodology / Decision and Safety Flow

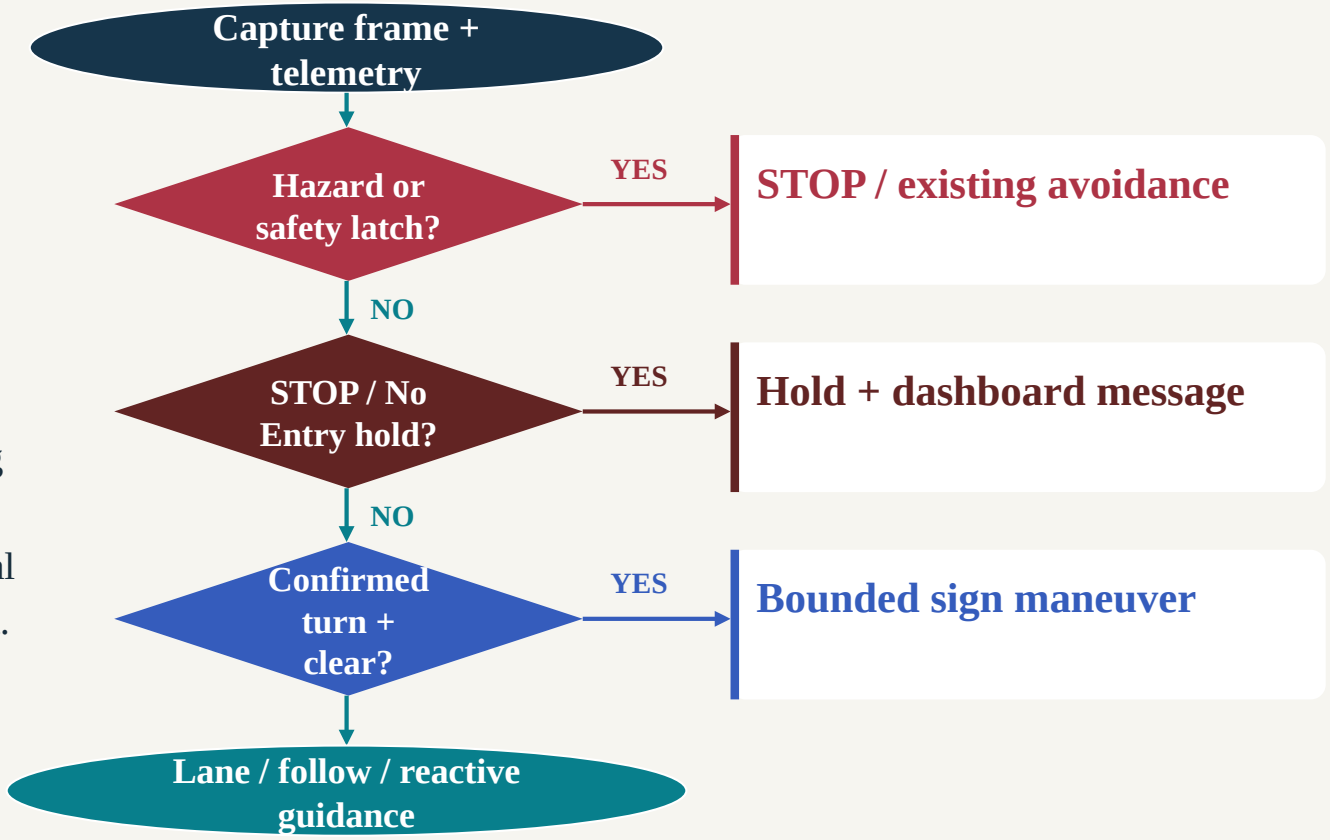
ARBITRATION / ONE COMMAND OWNER AT A TIME

OBSERVE

Only fresh sensing can authorize a sign maneuver.

RE-EVALUATE

Interruption during a sign turn → stop and fault. Uno local safety can preempt.

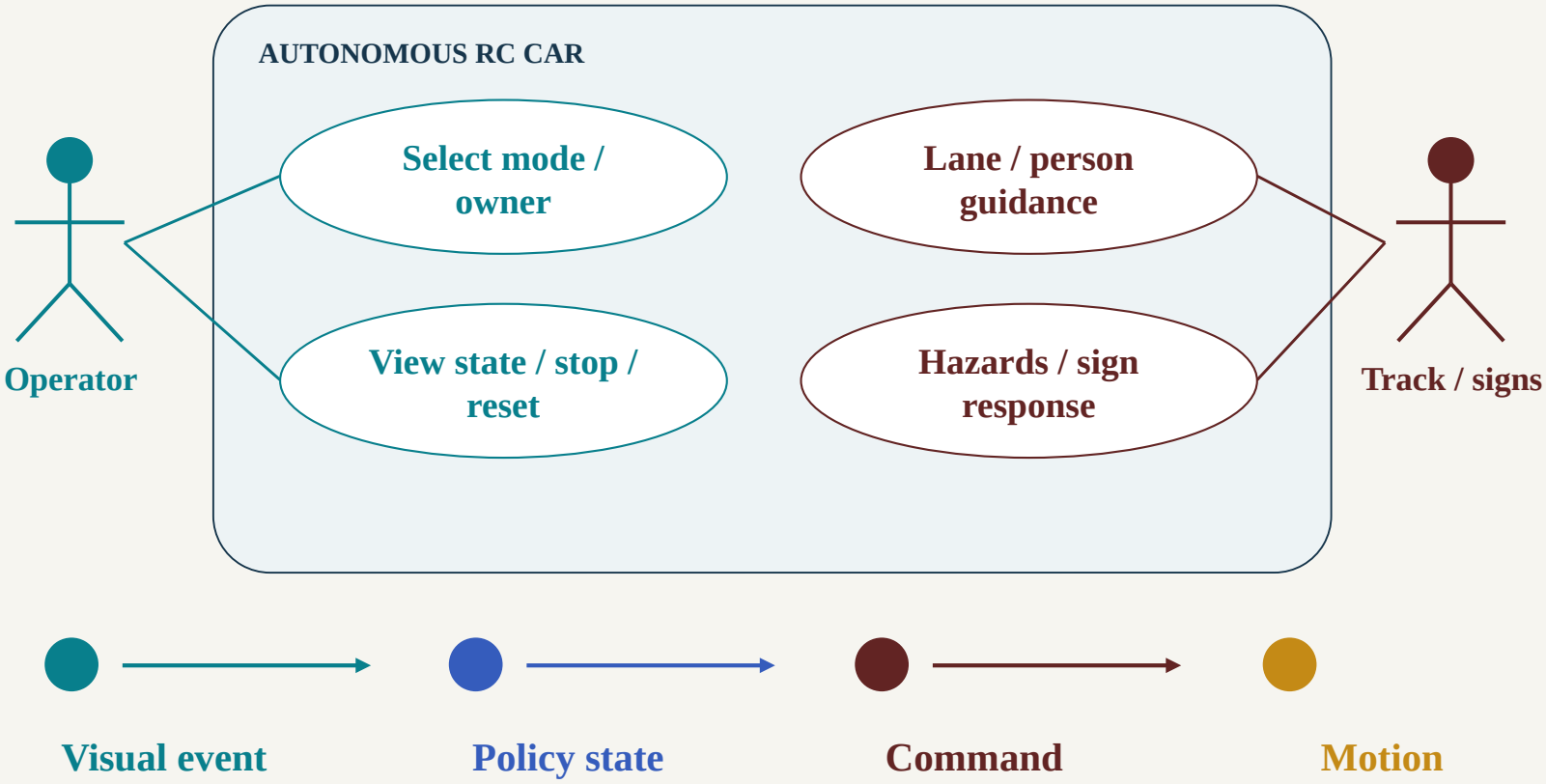


Sign extension: observations ≤0.75 s old; sonar ≤0.8 s; obstacle results ≤0.5 s.

Sources: car/main.py; Uno motion FSM; Traffic Sign/car/main.py and TrafficSignPolicy (separate trained extension).

Use Cases / From Operator to Outcome

BEHAVIOR CONTRACT / WHAT THE USER ASKS AND THE CAR MUST DO

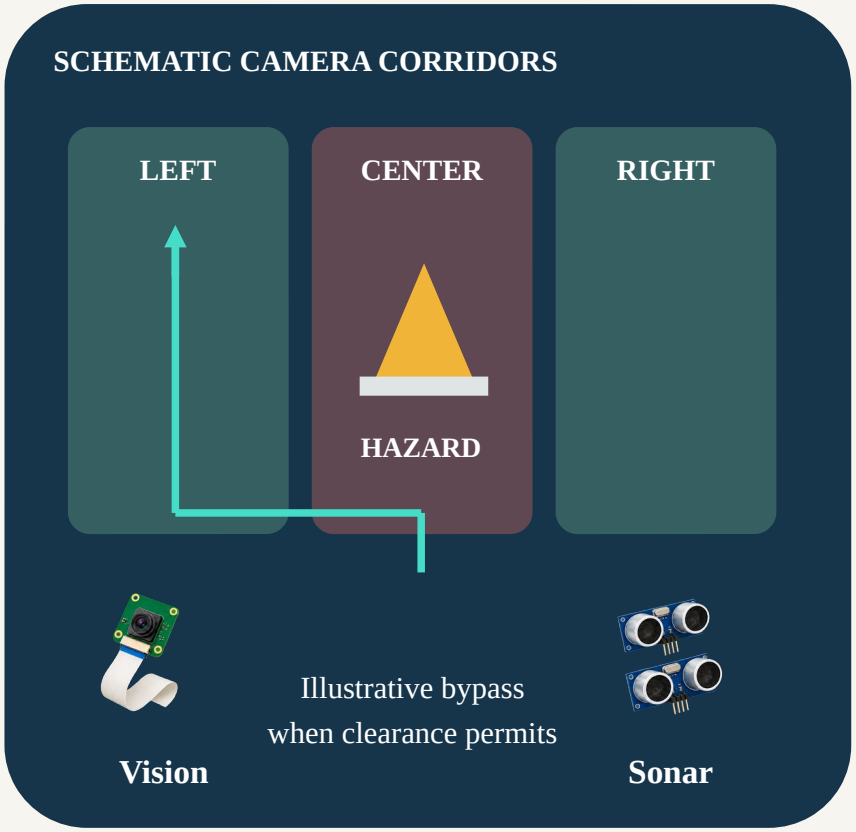


A complete demonstration pairs the detected event with policy state, commands and observed movement.

Sources: repository dashboard routes and operating modes; trained sign policy; conceptual UML diagram.

Obstacle Detection / Avoidance Logic

OBSTACLE AVOIDANCE / VISUAL HAZARDS + FRONT SONAR → MOTION PRIORITY



Detect candidate hazards

YOLOv10n object boxes join person and pothole detections.

Assess the path

Score left, center and right using box geometry, confidence and front sonar.

Choose the motion

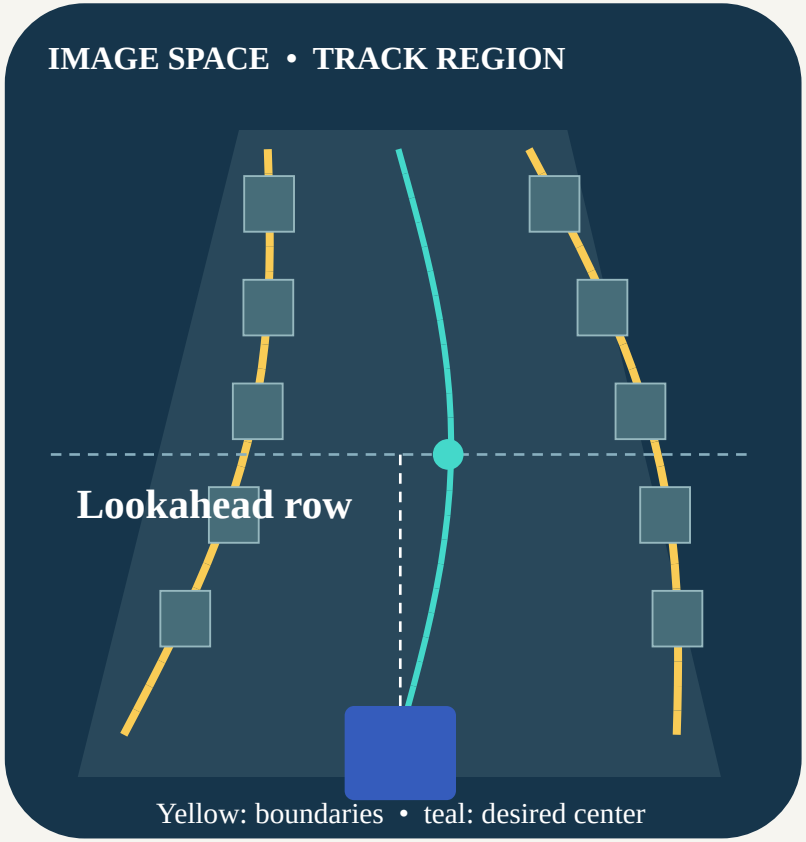
Close central danger → STOP; otherwise slow or bypass toward a clearer side.

Safety arbitration takes priority over normal lane / follow guidance; Uno reflex states retain control.

Sources: car/main.py _check_obstacle_ahead and mode handlers; perception/avoidance.py, Krishna main 9aeb238. Drawing is conceptual.

Lane Detection / Geometry to Steering

CLASSICAL VISION / TRACK STRUCTURE WITHOUT ANOTHER NEURAL MODEL



1 Mask + isolate

Mask lane colors in the lower-frame region.
Exclude overlapping hazard boxes.

2 Fit + stabilize

Sliding-window pixel search → robust curves.
Smooth across frames; reject sudden jumps.

3 Guide + recover

Estimate center from both lines or one.
Guide motion; prolonged lane loss → STOP.

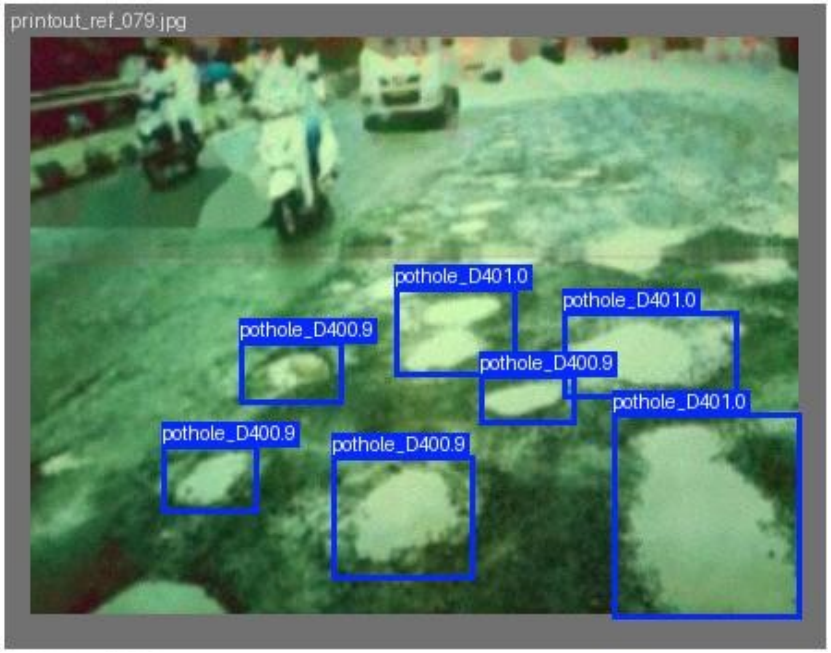
Steer toward the lane center while respecting the path direction.

Lane geometry supplies direction; confidence and loss handling determine when guidance can continue.

Sources: lane_detector.py and config.py, Krishna main 9aeb238. Diagram is illustrative image-space geometry, not a captured test.

Pothole Detection / Learning and Decisions

FROM VISUAL HAZARD TO RESPONSE / A MODEL OUTPUT YOU CAN EXPLAIN



Actual logged prediction on a printout

Detect road damage

D40 model adapted to printed road-damage scenes.

Filter candidate boxes

Apply confidence and lower-image checks to accept relevant boxes.

Choose a safe response

Feed boxes into stop / bypass logic; sonar adds proximity information.

Road-damage and printout validation are different domains; their logged scores are shown together on the results slide.

Sources: detector.py; printout_gpu_pothole_model/val_batch0_pred.jpg, Krishna main 9aeb238. This is logged model output, not track-test proof.

Traffic Signs / Train → Confirm → Act

FOUR-CLASS TRAINED EXTENSION / DELIVERED SEPARATELY FROM KRISHNA MAIN



STOP

Stop dwell: 3 s



NO ENTRY

Hold until reset



LEFT

Bounded left turn



RIGHT

Bounded right turn

TRAINING / ACTUAL ARTIFACTS

Four road-sign classes
Varied synthetic scenes
Fine-tuned YOLOv10n
Lightweight NCNN deployment

1. **Confirm across fresh frames**
   Reject stale or weak predictions
2. **Check size and image position**
   Act only when the sign is suitably visible
3. **Clearance → bounded action**
   Lane mode: bounded pulses + reacquisition

Camera frame → sign recognition → safety checks → action and dashboard feedback.

Sources: Traffic Sign/MODEL_CARD.md, heldout_summary.json, action_check_ncnn.json and main.py / traffic_sign_policy.py.

Result Analysis / Scores and Verification

MODEL METRICS / MEASUREMENTS AND VALIDATION SCOPE

Evaluation domain	Precision	Recall	mAP50	mAP50–95
RDD2022 road damage	0.53992	0.27092	0.25226	0.11421
Pothole printout run	0.99902	0.95506	0.97333	0.70171
Traffic signs: synthetic	0.98462	0.99896	0.99486	0.97783

WHAT IS VERIFIED

- Logged validation scores and training curves
- Four sign image-to-command outcomes
- Lane algorithm and software thresholds

WHAT NEEDS TRACK MEASUREMENT

- Lane error / route completion rate
- Camera-to-motion delay and braking distance
- Sustained fresh-inference FPS on Pi

Loop FPS ≠ fresh detection FPS. General neural detection runs every third loop iteration.

Sources: GitHub CSVs at 9aeb238; separate sign extension validation JSON; no dated physical Pi performance log supplied.

Conclusion and Future Scope

DELIVERED CAPABILITY / A CLEAR PATH TO STRONGER PHYSICAL EVIDENCE

Pi

Separate workloads

Pi for perception; Uno for actuation via an explicit serial command interface.

CV

Fuse complementary cues

Lane geometry and learned hazards, complemented by nearby-obstacle sonar.

UI

Make actions observable

Signs and motion states appear as dashboard feedback for observable decisions.

NEXT EXPERIMENTS

Measure

Latency + braking

Calibrate

Turn + camera geometry

Generalize

Real photos + rear sensing

Next hardware step: encoders / completed IMU feedback for measured motion instead of timed estimates.

Sources: final report conclusion and limitations; current source code; trained sign extension reports.

SDG Mapping / Engineering with a Purpose

PRIMARY ALIGNMENT: SDG 9 / SUPPORTING LINKS: SDG 3 AND SDG 11

9

INDUSTRY, INNOVATION & INFRASTRUCTURE

Target 9.5

Affordable robotics research. Modular code and shared validation artifacts.

Direct research / innovation link

3

GOOD HEALTH AND WELL-BEING

Target 3.6

Study obstacle sensing and stop decisions as foundations for collision-risk research.

No injury-reduction claim

11

SUSTAINABLE CITIES AND COMMUNITIES

Target 11.2

Explore safer mobility technology through a small controlled-track platform.

No deployed transport claim

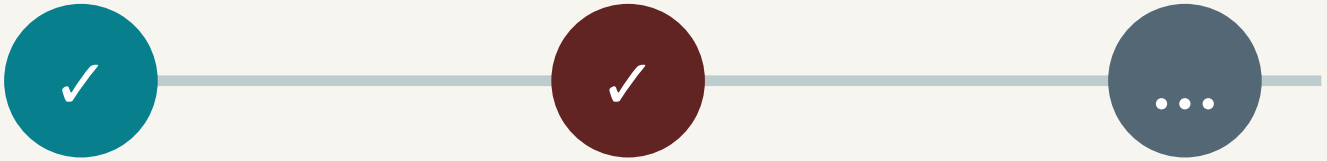
Primary: SDG 9.5 research and innovation. Supporting: safety and mobility research (3.6 and 11.2).

Sources: official UN targets at sdgs.un.org/goals/goal9, /goal3 and /goal11. Potential contribution, not measured societal impact.

Paper Status Till Final Presentation

PUBLICATION PROGRESS / MANUSCRIPT → SUBMISSION → DECISION

Autonomous RC Car Using Two-Layer Edge Perception and Real-Time Embedded Control



Manuscript prepared

SUBMITTED

Decision awaited

Venue and decision details will be added when available.

Current team-confirmed status: submitted. Acceptance has not been confirmed.

Source: team confirmation in this chat; supplied final manuscript. Venue, submission ID and decision email were not provided.

References / Reproducible Evidence

RESEARCH → CODE → MODEL LOGS / CLICKABLE SOURCES

METHODS

[1] Wang et al. YOLOv10: Real-Time End-to-End Object Detection. 2024. arXiv:2405.14458.

[2] Henriques et al. High-Speed Tracking with Kernelized Correlation Filters. IEEE TPAMI, 2015. DOI:10.1109/TPAMI.2014.2345390.

[3] Arya et al. RDD2022: A multi-national image dataset for automatic road damage detection. 2024. DOI:10.1002/gdj3.260.

[4] Müller & Koltun. OpenBot: Turning Smartphones into Robots. ECCV, 2020.

[5] Paull et al. Duckietown: An Open, Inexpensive and Flexible Platform for Autonomy Education and Research. ICRA, 2017.

IMPLEMENTATION & EVIDENCE

[6] Tencent NCNN; OpenCV official framework and tracking documentation.

[7] menonkrishna57/model-car-capstone, GitHub main 9aeb238. Runtime, firmware and both runs_rdd/results.csv files.

[8] Supplied final paper and 68-page report; separate Traffic Sign model card, synthetic evaluation and command-check reports.

[9] United Nations: official Sustainable Development Goals 3, 9 and 11 and associated targets.

Results-table row labels link directly to the logged metric sources.

Repository review basis: Krishna main 9aeb238. Detailed traceability and examiner Q&A are in the speaker notes and companion guide.